ROBOTICS PROJECT · SPRING 2026 - TEAM 1

Cleaning & Organizing Robot

An Imitation Learning Approach Inspired by AlohaMini

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01 CORE INSPIRATION

AlohaMini Framework

Low-cost hardware design enabling imitation learning

High-dexterity manipulation via dual-arm teleoperation

Imitation learning from human demonstrations — no RL needed

Our Adaptation

SO101 arm — single stationary unit

Dual-camera setup — overhead depth + wrist

Adapted to **cleaning & organizing tasks** within fixed work with **Imitation learning (ACT Policy)**

02 UPDATED PROJECT DIRECTION

Original Concept

General task execution using
AlohaMini-style setup

→

PIVOT

Current Direction

Cleaning & Organizing Robot
Focused on table/desk surfaces

Why the Pivot?

Instructor restriction: The LeRobot setup must remain stationary for the entire task duration.
Solution: Maximize the SO101 arm's work envelope from a single fixed base point — cleaning/organizing tasks are well-suited to this constraint.

03 TASK DEMO: FRUIT WIPING

One fruit at a time — bimanual coordination with tool use

① Grab

Simultaneous

- L — Grabs fruit from left bucket
- R — Grabs the towel

→

② Wipe

Both arms lift

- L — Holds fruit steady
- R — Wipes towel across fruit

→

③ Place & Return

Final position

- L — Places clean fruit in right basket
- R — Returns towel to original spot

L = Left Arm (fruit)   R = Right Arm (towel)

04 TECHNICAL & MITIGATION

Limitation

Gripper has limited flexibility, making consistent wipe motion a challenge — addressed through careful demonstration style during teleoperation.

Dual-Camera Setup

Issue: 3D spatial awareness is critical for stacking and occlusion handling. **Mitigation:** Overhead depth camera for global scene + wrist-mounted camera for fine manipulation.

Imitation Learning

Issue: Collecting sufficient demonstrations within constrained environment. **Mitigation:** 30 teleoperation episodes · 100,000 ACT training steps · 3+ hours on RTX 5090.

30 Teleoperation Episodes

100K ACT Training Steps

3 hrs Training Time (RTX 5090)

05.0 FUTURE MILESTONES

1

Perception Improvement

Identify object shape, cloth position, and scene layout more reliably.
Estimate graspable regions, object orientation, and possible collision risks in cluttered environments.

2

Candidate Action Design

Generate and compare multiple grasping and wiping poses.
Evaluate which candidate is safer, more stable, and more suitable for cleaner wiping results.

3

Task-Level Planning

Add high-level task logic for multi-step manipulation.
Track whether the cloth is still in hand, whether the object needs re-grasping, and when the robot should continue wiping or recover the cloth.

05.1 EVALUATION GOALS

Our target parameters — what success looks like for this project

✓ Task Success Rate

Target: fruit successfully placed in right basket across multiple runs.

↕ Grasp Success

Target: both arms reliably grasp their objects on each attempt.

~ Wipe Execution

Target: towel makes consistent contact and completes the wipe motion.

↩ Reset Accuracy

Target: towel returns to starting position after each cycle.

T Cycle Time

Goal: complete the full grab → wipe → place → reset cycle in under 15 seconds.

G Generalization

Goal: policy works even when fruit is slightly off its expected position.

MEET THE TEAM

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Teleoperation · Training · Evaluation

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Thank You

Questions welcome!